

Discussion #3W

Note: Your TA will probably not cover all the problems. This is totally fine; the worksheets are not designed to be finished within an hour. They are deliberately made slightly longer so they can serve as resources you can use to practice, reinforce, and build upon concepts discussed in lectures and labs.

Pandas Practice

The following questions are based on the elections.csv data referenced in the lecture. Assume that you have run the loaded data into the elections variable as follows:

```
elections = pd.read_csv('elections.csv')
```

	Year	Candidate	Party	Popular vote	Result	%
0	1824	Andrew Jackson	Democratic-Republican	151271	loss	57.210122
1	1824	John Quincy Adams	Democratic-Republican	113142	win	42.789878
2	1828	Andrew Jackson	Democratic	642806	win	56.203927
3	1828	John Quincy Adams	National Republican	500897	loss	43.796073
4	1832	Andrew Jackson	Democratic	702735	win	54.574789
...	...	...	...	...	...	...
182	2024	Donald Trump	Republican	77303568	win	49.808629
183	2024	Kamala Harris	Democratic	75019230	loss	48.336772
184	2024	Jill Stein	Green	861155	loss	0.554864
185	2024	Robert Kennedy	Independent	756383	loss	0.487357
186	2024	Chase Oliver	Libertarian Party	650130	loss	0.418895

187 rows x 6 columns

- Write a line of code that returns the elections table sorted in descending order by "Popular vote". Store your result in a variable named sorted.

Would calling `sorted.iloc[[0], :]` give the same result as `sorted.loc[[0], :]`?

- Using Boolean slicing, write one line of pandas code that returns a DataFrame that only contains election results from the 1900s.

3. Write one line of pandas code that returns a Series, where the index is the "Party", and the values are how many times that party won an election. Only include parties that have won an election.

4. Write a line of pandas code that returns a Series whose index is the years and whose values are the number of candidates that participated in those years' elections.

5. Write a line of pandas code that creates a filtered DataFrame named filtered parties from the elections dataset and keeps only the parties that have at least one election % more than 50%.

6. Write a line of pandas code that uses the filtered partiesDataFrame to return a new DataFrame with row indices that correspond to the year and columns that correspond to each party. Each entry should be the total percentage of votes for all the candidates that ran during that particular year for the specified party. Missing values from the dataset (the cases where a party did not have a candidate in a particular year) should be entered as 0. Below is an example.

Party	Democratic	Democratic-Republican	National Union	Republican	Whig
Year					
1824	0.000000	100	0.000000	0.000000	0.000000
1828	56.203927	0	0.000000	0.000000	0.000000